

Course Title

Intro to Machine Learning

Semester

Spring 2025

Class Time and location

TR 3:30 – 4:45pm

CSC 461

INSTRUCTOR**Name:** Dr. Nadine Abbas**Email:** nadine.abbas@lau.edu.lb**Office:** Nicol Hall 518**Office Hours:** TR 12:30 pm – 2 pm (or by appointment)**Webex Link:** <https://lau.webex.com/meet/nadine.abbas>**CURRENT CATALOG DESCRIPTION**

This course provides an overview of theoretical and application aspects of machine learning. Topics include supervised and unsupervised learning including generative/discriminative learning, parametric/non-parametric learning, neural networks, support vector machines, clustering, dimensionality reduction, and kernel methods. The course also covers learning theory, reinforcement learning, adaptive control. An applied approach will be used, where students get hands-on exposure to ML techniques through the use of state-of-the-art machine learning software frameworks.

COURSE PREREQUISITE/CO-REQUISITE*CSC310: Algorithms and Data Structures***TEXTBOOK AND REFERENCES**

- "Hands-On Machine Learning with Scikit-Learn and TensorFlow", Aurélien Géron, O'Reilly, 2017, ISBN: 978-1-491-96229-9
- "Evaluating Machine Learning Models", Alice Zheng, O'Reilly, 2015, ISBN: 978-1-491-93246-9
- "Python Machine Learning Projects", Brian Boucheron and Lisa Tagliaferri, Digital Ocean, 2019, ISBN: 978-0-9997730-2-4
- "A Hands-On Introduction to Machine Learning", Chirag Chah, Cambridge University Press, 2023 ISBN 978-1-009-12330-3
- "Data Mining: Concepts and Techniques", Jiawei Han Jian Pei Hanghang Tong, Morgan Kaufmann Elsevier, 4th Edition, 2023, ISBN: 978-0-12-811760-6
- "An Introduction to Statistical Learning with Applications in Python", Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor, Springer, 2023

COURSE TYPE

Required

☐

Elective

☒

Selective Elective

☐**COURSE LEARNING OUTCOMES**

- CLO.1 Describe the most important algorithms and the theory that constitutes the basis for machine learning
- CLO.2 Explain the principle for machine learning and how the algorithms and the methods can be used
- CLO.3 Discuss advantages with and limitations of machine learning for different applications
- CLO.4 Build basic intelligent systems using Python programming language and Scikit-Learn library

TOPICS COVERED IN THE COURSE

- Introduction to Machine Learning
- Refresher on Python Programming
- Classification and Regression
- Clustering
- Training Models
- Support Vector Machine (SVM)
- Regularization: Ridge and Lasso Regression
- Decision Tree
- Ensemble Learning
- Dimensionality Reduction
- Reinforced Learning
- Evaluating Machine Learning Models
- Linear discriminant analysis
- Generalized Additive Models
- Real-World Applications of Machine Learning

COURSE GRADING AND PERFORMANCE CRITERIA (Subject to 5% variation)

- Assignments, Attendance and Participation: 15%
- Midterm: 25%
- Final Exam: 35%
- Research project: 25%

POLICY ON CHEATING AND PLAGIARISM

Students caught cheating on an exam receive a grade of zero on the exam in their first cheating attempt and receive a warning. Students caught cheating for the second time will receive a grade of "F" in the course and another warning. Plagiarism on assignments and project work is a serious offense. If plagiarism is detected, a student will be subject to penalty, similar to the cheating case, which ranges from receiving a zero on the assignment concerned to an "F" in the course in addition to a warning.

UNIVERSITY ATTENDANCE POLICY

1. Students are expected to attend all classes.
2. For valid reasons, students may miss classes for a maximum equivalent to two regular weeks.
3. A student who misses more than the equivalent of two weeks of instruction in a particular course in a regular semester, or one week in a summer session, excused and otherwise, for any reason, is advised to withdraw from the course within the official withdrawal deadline. Otherwise, a grade of F will be assigned.
4. When exceeding the maximum number of absences, it is the instructor's prerogative to ask the concerned student to stop attending and drop the course. In this case, it is the student's responsibility to drop the course, otherwise a grade of "F" or "NP" will be given.
5. In exceptional justified cases (long illness, etc...), where absences exceed the maximum, the student has to petition to the department Chair to be allowed to stay in the course.
6. Students are held responsible for all the material presented in the classroom, even during their absence.

COURSE EVALUATION AND GRADE RELEASE

In order to improve the effectiveness of the educational process, all students are expected to submit their course evaluations by the last day of classes. Students who fail to complete the evaluation of ALL registered courses by the set deadline:

1. will not be able to access their course grades from Banner or Portal until two weeks after the end of the final exams period; and
2. will not be able to request transcripts.

The anonymity of the process and the students will be maintained at all times.

WITHDRAWAL POLICY

Students are advised to consult the University Official Policy regarding courses withdrawal at the following link: <http://www.lau.edu.lb/academics/arp/u/withdrawal-from-university.php>. In specific:

- WI (Early Withdrawal) Indicates withdrawal from the course, after the Late Registration Period and until the end of the 5th week of the Fall and Spring semesters, and until the 10th day of the Summer modules. It has no quality points. It does not count in the GPA, and no credits will be added to the student's record.

- WP (Withdrawal Pass) indicates withdrawal from the course, after the 5th week and until the end of the 10th week of the Fall and Spring semesters, and from the 11th day of classes until 18th day of the Summer modules. It has no quality points. It does not count in the GPA, and no credits will be added to the student's record.
- WF (Withdrawal Fail) indicates withdrawal from the course, after the 5th week and until the end of the 10th week of the Fall and Spring semesters, and from the 11th day of classes until 18th day of the Summer modules. It has no quality points. It does not count in the GPA, and no credits will be added to the student's record but is counted as repeat.

A **Withdrawal Form** must be submitted to the Registrar's Office.

Deadline for the "WP" and "WF" withdrawal from courses: check the university calendar
(It is the student's responsibility to drop the course)

STUDENT CODE OF CONDUCT – ACADEMIC VIOLATIONS

The following table defines the sanction(s) associated with each violation. In some cases, and when the violation is too general, a range of sanctions is set for the pertinent committee to choose from depending on the specifics of each case. As for the second offense, the set sanctions apply regardless of whether the violation has taken place in the same course or a different one, within the same semester or not.

Code #	Violation	First Offense	Second Offense
Cheating			
2.2.1	Using material or equipment (including mobile phones, electronic tablets, i-pads, calculators, and other devices) that is not authorized by the instructor in an examination, project, or graded assignment	zero on the deliverable with a warning	F on the course with a warning
2.2.2	Cheating, copying, collaborating with or aiding another Student in a manner not permitted by the instructor on an examination, project, or other graded assignment	zero on the deliverable with a warning	suspension
2.2.3	Distributing or aiding in the distribution of previous exams without authorization of the instructor	double warning – suspension	suspension – expulsion
2.2.4	Stealing, reproducing, or circulating an examination or other graded assignment before it has been administered	suspension	expulsion
2.2.5	Impersonating another Student or allowing another Student to impersonate one's self during an examination, presentation, or other graded assignment	suspension for both	expulsion
2.2.6	Impersonating an assistant, staff member, or faculty member for the purpose of (a) proctoring examinations without authorization or permission or (b) obtaining confidential information regarding coursework or examinations	suspension – expulsion	expulsion
2.2.7	Receiving, purchasing or selling a project, paper, or any academic document and presenting it as work other than that of the author	suspension – expulsion	expulsion
2.2.8	Submitting identical papers or coursework for credit in more than one class without the permission of the instructor	zero on the deliverable with a warning	F on the course with a warning
Plagiarism and Copyright Violations			
2.2.9	Failing to attribute language or ideas to their original source by not crediting the original author with an appropriate acknowledgement or citation	zero on the deliverable with a warning	F on the course with a warning
2.2.10	Using photocopied or electronic copies of textbooks, compact disks, films, music, online course materials, and other content beyond the fair use policy within University Premises	warning	double warning
2.2.11	Using copyrighted materials, including in written research reports and papers, without obtaining required permission, if any, from the rights holder	warning	double warning
Unauthorized Sale, Distribution, or Use of Course Materials			
2.2.12	Recording any lecture or presentation for personal use or public distribution without the prior consent	warning	double warning

Code #	Violation	First Offense	Second Offense
	of the course instructor. This applies to the unauthorized use of any medium including but not limited to mobile phones, electronic tablets, i-pads recorders, films, and other devices		
2.2.13	Selling academic materials by any Student, club, or group. This includes but is not limited to lectures, course recordings, class notes, and previous exams	warning	double warning

TIPS FOR SUCCESS

- **Attending the sessions is mandatory.**
- Deadlines for assignments and projects **must be respected**.
- Make-ups and Incomplete: students are not automatically entitled to make-ups; F will be given until reasons (in writing and within one week of absence) are presented and approved.
- You are responsible for all business conducted and announcements made during any scheduled class period, by email or posted over blackboard.
- Some exam questions will be based on class discussions, projects, and assignments.
- No mobile phones and laptops in the classroom and exams.